

ROJA ESWARAN

rojaeswaran@gmail.com | linkedin.com/in/roja-eswaran | rojaeswaran.github.io/bio

EDUCATION

Doctor of Philosophy in Computer Science Aug 2020 – Jun 2024

Binghamton University, Watson College of Engineering, USA

Dissertation: *"Sharing-Aware Live Migration of Virtual Machines"*

Advisor: Prof. Kartik Gopalan, Binghamton University

Master of Science in Computer Science, GPA: **3.8/4.0** Jan 2019 – May 2020

Binghamton University, Watson College of Engineering, USA

Courses: Operating Systems, Systems Programming in Rust, Computer Networks, Computer Architecture

Bachelor of Science in Computer Science and Engineering, GPA: **8.4/10** Aug 2014 – May 2018

Anna University, MEPCO Schlenk Engineering College, India

TECHNICAL SKILLS

Languages: C, Rust

Tools and Technologies: QEMU/KVM Virtualization, CI/CD, Kata Containers, Kubernetes, KubeVirt, OpenStack

PROFESSIONAL EXPERIENCE

Senior Engineer, GEICO, Raleigh, NC Dec 2024 – Present

- Produce, maintain, and patch high-quality, vulnerability-free container images for the application development team, and hardened VM images for the OpenStack Cloud team.
- Evaluated the feasibility and performance of several open-source VM-hardened container runtimes; presented findings in a talk titled *"A Tug-of-War: Performance vs. Security Tradeoff in Containers"* at **All Things Open (ATO '25)** in Raleigh, NC.
- Delivered a podcast titled *"Three Advantages: Why Edge Computing Demands a Different Developer Mindset"* at **All Things Open (ATO '25)** in Raleigh, NC.

Research & Development Engineer, Zededa Inc., San Jose, CA Aug 2023 – Dec 2024

- Enabled edge compute clusters with KubeVirt/EVE and resolved VM memory oversubscription in QEMU-KVM/EVE.

Research Assistant, Operating Systems and Networks Lab, Binghamton, NY Jan 2020 – Jun 2024

- Analyzed the performance of TCP- and RDMA-aware live migration on a Mellanox ConnectX-3 Pro network card.
- First-authored research article *"Tackling Memory Footprint Expansion During Live Migration of Virtual Machines,"* accepted at the **IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing (CCGrid '24)**.
- First-authored doctoral symposium article *"Incorporating Memory Sharing-Awareness in Multi-VM Live Migration,"* accepted at the **IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing (CCGrid '24)**.
- First-authored research article *"Template-aware Live Migration of Virtual Machines,"* accepted at **EdgeComm**, the **ACM/IEEE Symposium on Edge Computing (SEC '23)**.
- Authored research proposal *"Reducing Hypervisor Overhead for Virtual Interrupt Delivery in KVM/ARM Platform,"* approved for the **Research Proposal Exam (RPE)** at Binghamton University, Fall 2020.

File System Intern, Teradata Corporation, San Diego, CA May 2023 – Aug 2023

- Implemented gdb/file support for the Object File System (OFS) Spool in Teradata's proprietary VantageCloud.

Research Intern, Industrial Technology Research Institute (ITRI), Taiwan Jul 2022 – Dec 2022

- Added seamless, chained templating and migration features to the VM templating mechanism.

Research Intern, Argonne National Laboratory, Illinois Jan 2021 – May 2021

- Improved DGEMM parallelism using user-level threads.

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OPEN-SOURCE CONTRIBUTIONS

OpenStack / diskimage-builder

- Made bootstrapping three times faster by replacing debootstrap with mmdebstrap in diskimage-builder.
- Added DIB_MMDEBSTRAP_EXTRA_ARGS support to allow mmdebstrap to be used with custom arguments while preserving debootstrap compatibility.
- Co-authored a fix to disable the broken centos/10-stream-build-succeeds functional test caused by an empty upstream image hash file.

LF Edge / EVE

- Integrated new QEMU CPU-pinning feature into EVE.
- Reduced idle KubeVirt/EVE memory usage from 1.1 GB to 350 MB.
- Upgraded Xen tools from 4.18.0 to 4.19.0 for EVE to take advantage of new QEMU features.

Live Migration Performance Tooling

- *Memory-region-aware-copy*: optimizes traditional *memcpy* by copying only valid data instead of holes, reducing copy time tenfold.
- *Accurate QEMU timestamps*: measures total migration time and downtime of live migration.
- *QEMU overhead*: measures memory allocation overhead introduced by QEMU.
- *IOCTL interface*: retrieves the page frame number for a virtual page number using `/proc/pid/pagemap`.

Interrupt & Latency Research

- *Userfaultfd-aware-Mmap*: provides fine-grained control over a process's memory regions.
- *Kernel-level timer and inter-processor interrupt tests*: measure physical and virtual interrupt latency.

TECHNICAL PROGRAM COMMITTEE

- Reviewer, *Internet of Things and Cloud Computing* journal